

A faulty feeder selection method based on the slope of zero sequence current fitting curve for the non-effectively grounded system

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ABSTRACT

When a single-phase grounded (SPG) fault occurs in the distribution network, it is essential to ensure the distribution network's safe and reliable operation to select faulty feeders. This paper analyzes the slope of the zero sequence current fitting curve of each feeder when an SPG fault occurs in the distribution network. According to the different slope values between the faulty feeder and the healthy feeder at the zero sequence current mutation time, this paper proposed an SPG faulty feeder selection method based on the slope of the zero sequence current fitting curve. A locally weighted linear regression (LWLR) algorithm is first used to fit the zero sequence current sampled data. Then, the mathematical expression of the zero sequence current fitting curve is derived, and the corresponding sloping curve is obtained. Finally, the slope at the zero sequence current mutation time is used to determine the faulty feeder. If the slope value of all fitting curves at the zero sequence current mutation time is negative, the bus fault is determined; otherwise, the feeder with a positive slope value of the fitting curve is determined as the faulty feeder. In this method, the power frequency component of zero sequence current does not need to be filtered out, making it easy to achieve good detection sensitivity in high-resistance grounded fault selection. MATLAB/Simulink simulation results indicate that the proposed method can detect the SPG fault accurately under various fault conditions.

1. Introduction

In China, the low current grounded mode is applied widely in the mid-voltage distribution system [1]. The system is allowed to operate for 1~2 h since the phase angle and amplitude of the phase-to-phase voltage remains unchanged after an SPG fault occurs [2]. Nevertheless, it may cause the phase-to-phase grounded fault and enlarge the fault range if the SPG fault is not timely eliminated [3]. Presently, the distribution network topology is more complex due to the access to high-density distributed generation: power flow distribution and voltage distribution are changed, resulting in low feeder selection accuracy of the faulty feeder selection device [4]. Therefore, it is significant to analyze how to rapidly determine and remove the faulty feeder to ensure the non-effectively grounded system (NUGS) in safe and reliable operation [5].

According to different fault characteristic quantities used, the SPG faulty feeder selection methods in the NUGS are divided into steady-state and transient methods [6]. The steady-state methods include the zero sequence reactive power direction method [7], zero sequence

active power method [8], and harmonic current method [9], etc. The transient methods include the transient coulomb method [10] and the traveling wave method [11], etc. The steady-state methods require that the grounded arc is stable and the grounded resistance is constant, while there are intermittent grounded faults in the grounded faults [12]. Compared with steady-state fault characteristic quantity, transient zero sequence current amplitude is larger [13]. Consequently, the transient methods are more sensitive and adaptive in selecting the SPG faulty feeder [4]. However, these methods require a higher sampling frequency when selecting the faulty feeder.

Recently, faulty feeder selection methods based on current waveform morphology analysis have become increasingly common. A faulty locating method based on the slope of the current waveform for the medium voltage flexible DC distribution system is proposed in reference [14]. This method is concise in principle and simple in the calculation, but it only applies to the DC distribution system. Reference [15] proposed a method based on the sloping curve of the zero sequence current waveform for the AC distribution network to select the faulty feeder. Nevertheless, the power frequency component must be filtered out. Both

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references [14] and [15] highlight the fault characteristic and effectively improve the accuracy of faulty feeder selection. However, a better ability to collect and process data from the device is required in these methods.

Therefore, this paper analyzes the slope of each feeder's zero sequence current waveform. According to the different slope values of each feeder at the zero sequence current mutation time, an SPG faulty feeder selection method based on the slope of the zero sequence current fitting curve is proposed. This method performs the LWLR for the sampled data, then derives the fitting curve and uses the slope values of the fitting curve as a criterion for the faulty feeder selection. The simulation results and field test data under various fault conditions prove the effectiveness of the proposed method.

2. Analysis of the SPG fault characteristics of NUGS

When an SPG fault occurs in the NUGS, the transient characteristic information is richer, the phase voltage of the fault phase reduces to zero, the amplitude of that of the healthy feeder will increase by $\sqrt{3}$ times, and the zero sequence voltage of the fault spot is the sum of the triphasic ground voltage [16]. The equivalent circuit of the SPG fault transient analysis for NUGS is shown in Fig. 1 [17].

In Fig. 1, $e(t)$ is the voltage of the power supply, L_p is the inductance of the arc suppression coil, C_0 represents the differential value between the zero sequence distributed capacitance of the whole system and that of the downstream of fault spot, L and R represent the equivalent inductance and resistance, respectively.

2.1. Neutral non-grounded system

In Fig. 1, the equivalent circuit of a neutral non-grounded system is shown when K is disconnected, and the corresponding differential equation is as follows:

$$Ri_C + L \frac{di_C}{dt} + \frac{1}{C_0} \int i_C dt = U_{\phi m} \sin(\omega t + \varphi) \quad (1)$$

where i_C represents the capacitance current flowing through the feeder, $U_{\phi m}$ is the amplitude of $e(t)$. In Eq. (1), when $R < 2\sqrt{L/C_0}$, the system is underdamped, and the zero sequence current shows periodic oscillation attenuation characteristics; when $R > 2\sqrt{L/C_0}$, the system is overdamped, the zero sequence current shows a non-periodic oscillation attenuation characteristic.

When an SPG fault occurs in the NUGS, the i_C consists of the oscillation attenuation component i_{CI} and steady-state power frequency component i_{CII} , namely:

$$i_C = i_{CI} + i_{CII} = I_{Cm} \left[\left(\frac{\omega_f}{\omega} \sin \varphi \sin \omega t - \cos \varphi \cos \omega t \right) e^{-\delta t} + \cos(\omega t + \varphi) \right] \quad (2)$$

where $I_{Cm} = U_{\phi m} \omega C_0$ is the amplitude of capacitance current, $\delta = 1/\tau_c$

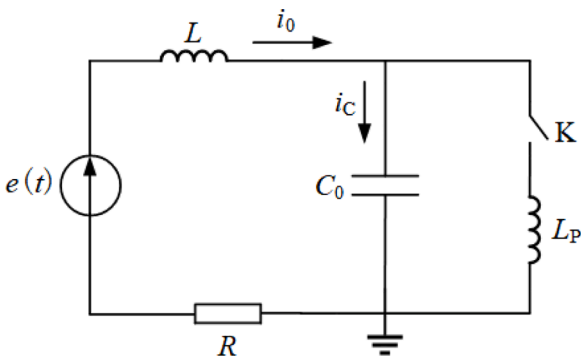


Fig. 1. Equivalent circuit for transient analysis of the SPG fault.

$= R/2L$ and $\omega_f = \sqrt{1/LC_0}$ represent the attenuation coefficient and the angle frequency of the free oscillation component, respectively. τ_c in the equation of $\delta = 1/\tau_c$ is time constant.

2.2. Resonant grounded system

When K is closed, the equivalent circuit of a resonant grounded system is shown in Fig. 1. The grounded current in the NUGS is composed of two components: the capacitance current and the inductance current. The transient capacitance current comprises the oscillation attenuation component i_{CI} and steady-state power frequency component i_{CII} , as shown in Eq. (2). If an SPG fault occurs at the moment of the voltage reaching the peak value ($\varphi = 90^\circ$), the amplitude of i_{CI} reaches the maximum in a quarter:

$$i_{CI\max} = I_{Cm} \frac{\omega_f}{\omega} e^{-\frac{T_f}{4T_c}} \quad (3)$$

In Eq. (3), $T_f = 2\pi/\omega_f$ is the period of free oscillation. The transient zero sequence current is

$$i_0(t) = i_C(t) = U_{\phi m} \omega C_0 \left(\frac{\omega_f}{\omega} e^{-\delta t} \sin \sqrt{\omega_f^2 - \delta^2} t - \sin \omega t \right) \quad (4)$$

In Eq. (4), $\sqrt{\omega_f^2 - \delta^2}$ is approximate to ω_f due to $\delta \ll \omega_f$, where ω_f is also known as the primary resonant angular frequency of the free component in $i_0(t)$. Therefore, the transient zero sequence current is approximated as follows:

$$i_0(t) = U_{\phi m} \omega C_0 \left(\frac{\omega_f}{\omega} e^{-\delta t} \sin \omega_f t - \sin \omega t \right) \quad (5)$$

When the SPG fault occurs, the direction of $i_0(t)$ of the faulty feeder is contrary to that of the healthy feeder at the first half cycle, which is called the transient stage. The transient zero sequence current amplitude is several times that of the steady-state zero sequence current. Therefore, the zero sequence current direction of the faulty feeder is contrary to that of the healthy feeder at the transient stage.

3. Faulty feeder selection based on the slope of zero sequence current fitting curve

3.1. Locally weighted linear regression

As a mathematical method of formulating discrete data, the linear fitting algorithm quickly occurs "underfitting" (errors caused by inaccurate fit data) and "overfitting" (errors caused by the excessive complexity of the modeling due to overfitting) [18]. The least squares fitting (LSF) algorithm used in reference [15] is inaccurate in predicting the waveform of the zero sequence current due to its minimum mean square error; such a case is called "underfitting". There is no "underfitting" problem when using the LWLR algorithm to predict the zero sequence current waveform; accordingly, we choose this algorithm to predict the zero sequence current waveform based on the sampled data.

The LWLR algorithm is an improved algorithm of general linear regression, which will assign weights to each prediction datum by introducing a weighting function. The weights of the sampled data in a particular region (containing more sampled data) are larger than that of the other data [19]. Therefore, compared with the general linear fitting algorithm, the LWLR algorithm can highlight the local characteristics of the sampled data and improve the fitting accuracy. Firstly, we need to find a cost function $F(\theta)$ to fit a new regression coefficient θ [20]. The cost function is defined as follows:

$$F(\theta) = \sum_{i=1}^m w_i (y_i - \theta^T x_i)^2 \quad (6)$$

where x_i is the i -th sample datum, y_i is the actual value, $\theta^T x_i$ is the predicted value, w_i is the weight of x_i . And θ is an initialized vector, the element of which can be all random numbers and will be updated in each

iteration as follows:

$$\hat{\theta} = \theta - \alpha \sum_{i=1}^m (y_i - \theta_i x_i) w_i x_i \quad (7)$$

where α is the learning rate determining the step length to move on in each iteration. The weight, w_i , is determined by the distance between the predicted data. The closer the distance, the greater the weight of x_i and the greater the influence of the corresponding error value ($f(x_i) - y_i$) on the fitting results. Therefore, adding polynomials to the polynomial linear fitting algorithm, the same result can be obtained by introducing weight w_i in the cost function. The weight, w_i , is computed as:

$$w_i = \exp\left(-\frac{(x_i - x)^2}{2\varepsilon^2}\right) \quad (8)$$

where x is the point to be predicted, ε is a parameter for controlling the changing rate of weight w_i . The larger the ε is, the faster the w_i decreases with the increasing distance between x_i and x ; therefore, the fitting results are affected by the ε . Extensive simulations show that the fitting results can best embody the distribution characteristics of the sampled data when $\varepsilon=0.001$. Accordingly, we determine that $\varepsilon=0.001$.

The simulation modeling of 10 kV NUGS is shown in Fig. 2. This system consists of three hybrid feeders (L_1 , L_2 , L_3) with overhead feeders and cables.

Fig. 2 shows a resonant grounded system when K is closed. Assume that an SPG fault occurs on L_1 , which will cause zero sequence current flowing in the NUGS. The LSF algorithm and the LWLR algorithm are used to fit the zero sequence current sampled data of L_2 , respectively. The fitting curve I_{sf} and I_{re} of the zero sequence current are shown in Fig. 3, which can visually show the superiority of the LWLR algorithm.

It can be seen from Fig. 3 that the I_{sf} is a straight line and cannot effectively show the changing trend of the zero sequence current, and such a case is "underfitting". By contrast, the I_{re} is more satisfactory than the I_{sf} , which can effectively prevent "underfitting".

3.2. Analysis of the slope of the fitting curve of zero sequence current

Suppose that a point x_0 and its neighborhood are all in the defined domain of the function $y = f(x)$. When x is increased by Δx at x_0 , and $(x_0 + \Delta x)$ is also in the neighborhood of x_0 , the corresponding function contains the increment $\Delta y = f(x_0 + \Delta x) - f(x_0)$ [21].

When $\Delta x \rightarrow 0$, if the limit of the ratio of Δy to Δx exists, the function $y = f(x)$ is derivable at point x_0 , and the limit value is the derivative of $y = f(x)$ at point x_0 , denoted as:

$$f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0) - f(x_0 - \Delta x)}{\Delta x} \quad (9)$$

It can be seen from Eq. (9) that $f'(x_0)$ is a locally linear approximation of $y = f(x)$, that is, it is the slope of the tangent of $y = f(x)$ at x_0 . Therefore, the $f'(x_0)$ can reflect the changing trend of y near x_0 . Taking

the derivative of Eq. (5), we obtain the first-order derivative expression of $i_0(t)$ as follows:

$$\frac{di_0(t)}{dt} = U_{\phi m} \omega C_0 \left(\frac{\omega_f^2}{\omega} e^{-\delta t} \cos \omega_f t - \frac{\delta \omega_f}{\omega} e^{-\delta t} \sin \omega_f t - \omega \sin \omega t \right) \quad (10)$$

If the first-order derivative of $i_0(t)$ is calculated directly, the fluctuation of $di_0(t)/dt$ is more significant. Therefore, the LWLR algorithm is first used to fit the sampled data to prevent this fluctuation. If an SPG fault occurs at L_1 , the zero sequence current sampled points and LWLR curves of L_1 , L_2 , and L_3 are shown in Fig. 4.

Fig. 4 shows that the amplitude of the fitting curve of L_1 is larger than that of other feeders. Moreover, the fitting curve of L_1 shows a remarkable increasing tendency near the fault time. In contrast, the fitting curve of the healthy feeder shows a decreasing trend near the fault time. Consequently, the fitting results can fully reflect the varying characteristics of the zero sequence current. Since the changing trend for the zero sequence current of the faulty feeder and the healthy feeder is opposite at the moment of failure, the polarity of the first derivative values of the zero sequence current are opposite.

The sloping curves of the zero sequence current fitting curve are obtained by deriving that of L_1 , L_2 , and L_3 , as shown in Fig. 5.

Fig. 5 shows that the slope value obtained for L_1 is positive near the fault time, whereas for the healthy feeder, it is negative. Therefore, we consider using the slope value of the I_{re} at the fault time to identify the faulty feeder. If the slope value of all fitting curves is negative, one will determine that a bus fault occurred; otherwise, the feeder with a positive slope value is determined as the faulty feeder.

3.3. Faulty feeder selection method principle

From the above analysis, we can see that the slope value of the I_{re} at the fault time is positive if an SPG fault occurs on a feeder, and its amplitude is large; if an SPG fault occurs on the bus, the slope value of that of every feeder is negative, and its amplitude is small. Accordingly, the above conclusion can be used for the SPG faulty feeder selection.

Based on the above analysis of fault characteristics, an SPG faulty feeder selection method in the NUGS based on LWLR is proposed in this paper. The slope of I_{re} at the zero sequence current mutation time is used as the fault criterion.

Assume that there are n feeders in the NUGS, the slope k_j of i_{jre} of the feeder j at the zero sequence current mutation time is as follows:

$$k_j = \frac{di_{jre}(t_0)}{dt_0} \quad (11)$$

If there is $k_j > 0$, that is, the slope value of the I_{re} near the zero sequence current mutation time is positive, the feeder fault is judged, and the fault occurs on the feeder j ; for all feeders, if $k_j < 0$, that is when the slope value of the I_{re} of each feeder is negative, the bus fault is determined.

The flowchart of the proposed method is shown in Fig. 6.

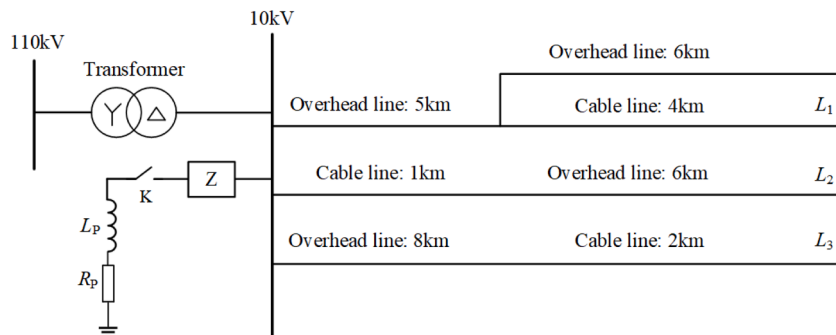


Fig. 2. The simulation modeling of NUGS.

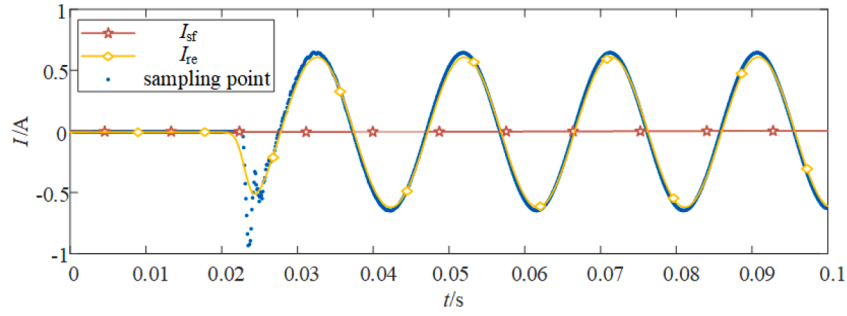


Fig. 3. The fitting curve of the zero sequence current of L_2 .

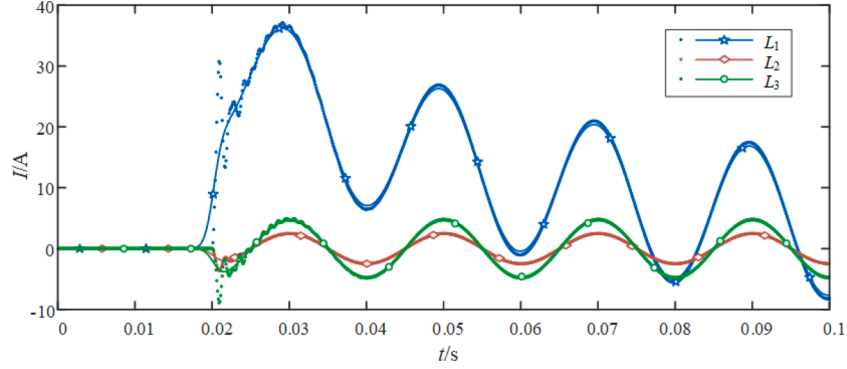


Fig. 4. The zero sequence current sampled points and their fitting curves.

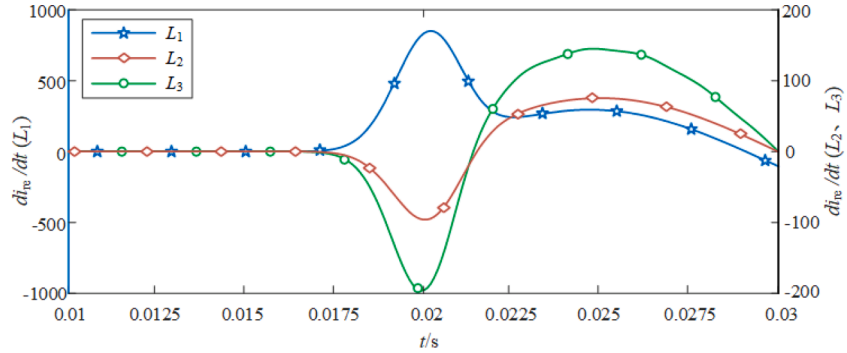


Fig. 5. The sloping curve of the zero sequence current fitting curve.

The detailed steps of the faulty feeder selection are as follows:

- (1) Collect zero sequence voltage U_0 .
- (2) Fault detection. When $U_0 > KU_N$ (U_N is the phase voltage, and K is usually between 0.1 and 0.15), the system is considered faulty, and the protection device starts up.
- (3) Extract fault characteristic signal. When a fault occurs in the NUGS, the zero sequence current of each feeder is directly measured by a zero sequence current filter composed of three current transformers, or the three-phase current of each feeder is measured and then calculated to obtain the zero sequence current.
- (4) The LWLR algorithm calculates the sampled data to obtain the I_{re} .
- (5) Calculate the slope of the fitting curve. By deriving the I_{re} , the slope k_j at the fault time of each feeder is obtained.
- (6) Faulty feeder selection. If $k_j > 0$, the feeder j is judged as the faulty feeder; if $k_j < 0$ is satisfied for each feeder, the bus fault is judged.

4. Simulation verification

The simulation modeling of 10 kV NUGS is established using the Simulink module of Matlab. The model consists of three hybrid feeders with overhead feeders and cable feeders (L_1, L_2, L_3), a cable feeder (L_4), and an overhead feeder (L_5), where $L_1=15$ km, $L_2=7$ km, $L_3=10$ km, $L_4=5$ km, $L_5=15$ km, respectively. The simulation model is shown in Fig. 7, and its parameters are shown in Table 1 [22].

The connection type of the transformer in Fig. 7 is Yd11, and the transformation ratio is 110/10.5 kV. The active power of the load connected to each feeder is unified to 1MW, the power factor is 0.9, the arc suppression coil compensation degree is 10%, and the sampling time interval is 1×10^{-4} s. Assume that an SPG fault occurs at the time of 0.04 s.

4.1. Simulation results and analysis

4.1.1. Different neutral grounded modes

The SPG fault that occurred in the NUGS is simulated under different

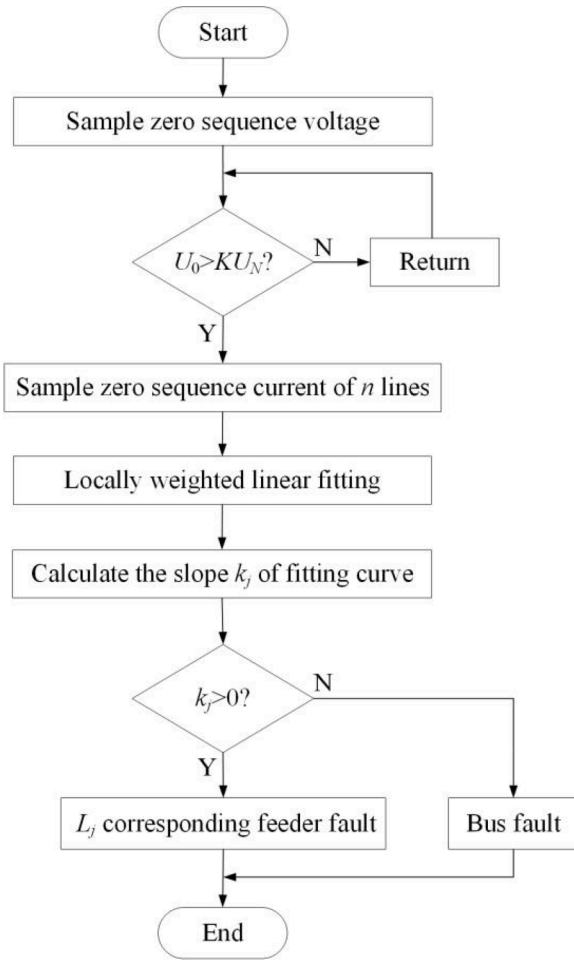


Fig. 6. The flowchart of the proposed method.

neutral grounded modes. If an SPG fault occurs at L_1 in a resonant grounded system, the distance between the fault spot and the bus is 3 km, the grounded resistance is 10Ω , and the fault inception angle is 0° . The sloping curve of the I_{re} of each feeder is shown in Fig. 8.

Fig. 8 shows that the slope value of the I_{re} of L_1 is positive near the fault time, whereas, for the healthy feeder, it is negative. Therefore, the slope k_j of the zero sequence current fitting curve near the fault time can directly reflect the fault characteristics. If k_j is positive, the L_j is determined as the faulty feeder; if $k_j < 0$, the L_j is selected as the healthy feeder.

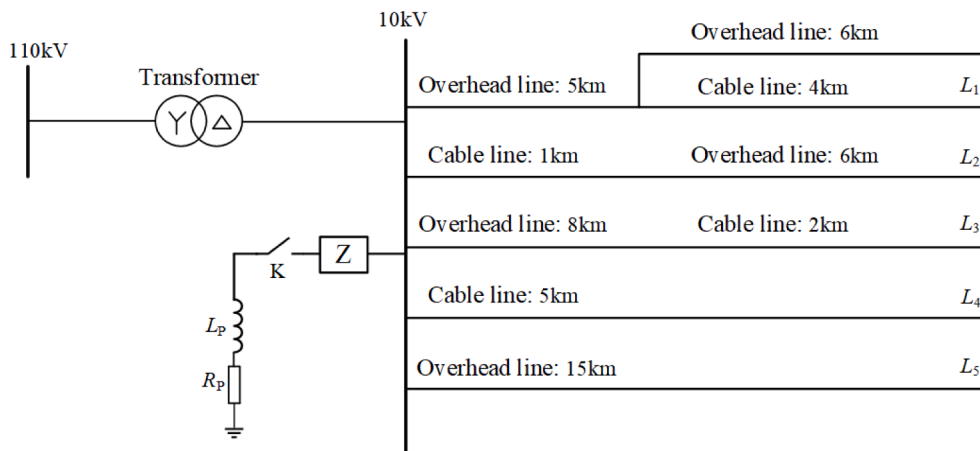


Fig. 7. The simulation model of NUGS.

The simulation results under different neutral grounded modes are shown in Table 2.

The simulation results in Table 2 show that when the simulation is carried out under different neutral grounded modes, the k_1 is positive and the amplitude is large, and k of L_2 , L_3 , L_4 , and L_5 is negative, and L_1 can be judged as the faulty feeder. Therefore, the neutral grounded modes do not affect the proposed method.

4.1.2. Different fault conditions

If an SPG fault occurs at L_2 , the distance between the fault spot and bus is 3 km, the fault inception angle is 0° , and the grounded resistance is 0Ω , 10Ω , 100Ω , 500Ω , 1000Ω , and 3000Ω , respectively. The simulation results are given in Table 3.

The simulation results in Table 3 show that with the increasing grounded resistance, the absolute value of k_j will gradually decrease. However, the k_2 keeps positive with the increasing grounded resistance. The k_1 , k_3 , k_4 , and k_5 are all negative, so L_2 is judged as the faulty feeder. The fault characteristic is still obvious as the grounded resistance increases to 3000Ω (high-resistance grounded fault). Therefore, the simulation results using the proposed method are not affected by the varying transition resistance.

If an SPG fault occurs on L_3 , the distance between the fault spot and bus is 5 km, the grounded resistance is 100Ω , and the fault inception angles are 0° , 30° , 45° , 60° , and 90° , respectively. The simulation results are shown in Table 4.

The simulation results in Table 4 show that the k_3 keeps positive, and the slope value of L_1 , L_2 , L_4 , and L_5 keep all negative with the increase of the fault inception angle. Accordingly, L_3 is judged as the faulty feeder. Therefore, the faulty feeder selection results using the proposed method are not affected by the varying fault inception angle.

If an SPG fault occurs on L_5 , the fault inception angle is 60° , the grounded resistance is 10Ω , and the distances between the fault spot and bus are 3 km, 5 km, 7 km, 10 km, and 15 km, respectively. The faulty feeder selection results are shown in Table 5.

The simulation results in Table 5 show that the slope value of L_5 keeps positive with the increase of distance between the fault spot and the bus, while the slope value of L_1 , L_2 , L_3 , and L_4 are all negative.

Table 1
The parameters for the simulation model.

Feeder type	Phase sequence	$R(\Omega/\text{km})$	$L(\text{mH}/\text{km})$	$C(\mu\text{F}/\text{km})$
Overhead feeder	Positive sequence	0.125	1.3	0.01274
	Zero sequence	0.275	4.6	0.0054
Cable feeder	Positive sequence	0.27	0.255	0.339
	Zero sequence	2.7	1.109	0.28

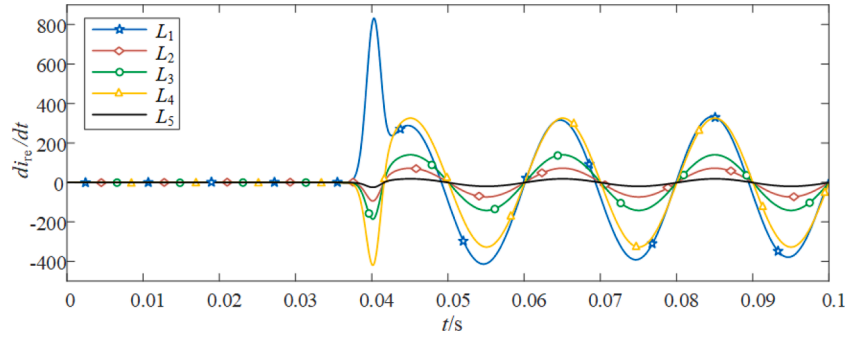


Fig. 8. The sloping curve of the zero sequence current fitting curve.

Table 2

The simulation results under different neutral grounded modes.

Neutral grounded mode	k_1	k_2	k_3	k_4	k_5	Results
Neutral point ungrounded	740.8	-95.8	-190.0	-430.1	-24.9	L_1
Resonant grounded	810.3	-94.0	-186.4	-422.0	-24.4	L_1

Table 3

The simulation results under different grounded resistances.

Grounded resistance/ Ω	k_1	k_2	k_3	k_4	k_5	Results
0	-364.0	1091	-182.2	-412.9	-23.8	L_2
10	-353.7	1050	-177.3	-402.2	-23.2	L_2
100	-210.6	616.6	-106.5	-244.0	-14.1	L_2
500	-66.1	193.3	-33.6	-77.4	-4.4	L_2
1000	-35.3	103.4	-18.0	-41.5	-2.4	L_2
3000	-12.3	36.1	-6.2	-14.5	-0.8	L_2

Table 4

The simulation results under different fault inception angles.

Fault inception angle/ $^\circ$	k_1	k_2	k_3	k_4	k_5	Results
0	-199.2	-51.8	530.3	-231.8	-13.4	L_3
30	-369.2	-92.2	1028.9	-414.5	-23.9	L_3
45	-437.6	-108.4	1209.3	-487.8	-28.2	L_3
60	-462.6	-113.7	1285.6	-512.3	-29.5	L_3
90	-419.6	-101.5	1180.0	-458.1	-26.4	L_3

Table 5

The simulation results under different fault distances.

fault distance/km	k_1	k_2	k_3	k_4	k_5	Results
3	-475.5	-108.5	-227.2	-494.5	1669.2	L_5
5	-620.6	-142.5	-298.3	-647.8	2053.3	L_5
7	-738.3	-172.3	-357.9	-780.8	2372.1	L_5
10	-842.7	-201.2	-412.7	-908.8	2656.6	L_5
15	-885.6	-216.9	-438.4	-976.8	2763.3	L_5

Accordingly, L_5 is determined as the faulty feeder. Therefore, the faulty feeder selection results using the proposed method are not affected by the varying distances between the fault spot and the bus.

To analyze the influence of the Gaussian white noise on faulty feeder selection, assuming an SPG fault occurs on L_1 , the fault inception angle is 30° , the distance between the fault spot and bus is 3 km, and the grounded resistance is 1000 Ω . Gaussian white noise with signal-to-noise ratios (SNRs) of 5 dB, 15 dB, 30 dB, 45 dB, and 60 dB is added, respectively. The simulation results are shown in Table 6.

From Table 6, we can see that the k_1 keeps positive with the increase of the SNRs, and the k_2 , k_3 , k_4 , and k_5 keep all negative after adding Gaussian white noise with different SNRs. Accordingly, L_1 is judged as the faulty feeder. Therefore, the faulty feeder selection results using the proposed method are not affected by the Gaussian white noise.

4.1.3. The influence of inverter-based distributed generation

To analyze the influence of the distributed generation (DG) connected to the distribution network on faulty feeder selection, two inverter-based DGs are added to the bus and L_4 , respectively. Assuming an SPG fault occurs on L_1 , the simulation results under different fault inception angles and grounded resistances are shown in Table 7.

From Table 7, we can see that the absolute value of k_j ($j = 1, 2, \dots, 5$) of the zero sequence current fitting curve in the distribution network with DG-connected is smaller than that of the distribution network without DG-connected. However, the k_1 keeps positive when an SPG fault occurs in the NUGS with DG-connected, and all the other slope values of k_2 , k_3 , k_4 , and k_5 keep negative. Accordingly, L_1 is judged as the faulty feeder. Therefore, the faulty feeder selecting results using the proposed method are not affected by the inverter-based DG access.

4.1.4. Bus fault

To analyze whether the proposed method can identify the bus fault correctly, it is assumed that the bus failure occurs in the NUGS under different fault inception angles and grounded resistances. The simulation results are shown in Table 8.

The k_j of each feeder is negative in Table 2. Accordingly, the bus fault is determined. Therefore, the proposed method can also determine whether a bus fault occurs.

4.2. Comparison with other methods

Three faulty feeder selection methods are compared with the proposed method to highlight the superiority of the proposed method. The transient reactive power direction method is the first method, where the transient reactive power Q_m is the average power of the Hilbert transform of U_0 and i_0 [23]. The second method is based on the distribution of zero sequence current, in which the faulty criterion deviation Sk is the measurement of the deviation direction and degree of zero sequence current distribution, that is, the asymmetric degree of transient zero sequence current sampling data distribution [24]. The third method is the amplitude comparison method of zero sequence current, where I_{m0} is

Table 6

The simulation results under Gaussian white noise with different SNRs.

SNR/dB	k_1	k_2	k_3	k_4	k_5	Results
5	38.3	-2.3	-4.0	-10.0	-0.6	L_1
15	32.8	-3.4	-6.2	-15.0	-0.9	L_1
30	30.7	-3.8	-7.0	-16.8	-1.0	L_1
45	30.3	-3.9	-7.1	-17.2	-1.0	L_1
60	30.3	-3.9	-7.2	-17.2	-1.0	L_1

Table 7

The simulation results under different fault conditions and with/without DG-connected.

Condition	Grounded resistance/ Ω	Fault inception angle/ $^\circ$	k_1	k_2	k_3	k_4	k_5	Results
/	0	0	857.6	-98.3	-195.8	-441.8	-25.6	L_1
DG			429.8	-49.0	-97.4	-220.5	-12.7	
/	10	90	831.6	-55.9	-123.0	-259.6	-14.5	L_1
DG			391.8	-23.3	-52.7	-110.8	-6.0	
/	100	60	1043.4	-111.6	-224.7	-502.9	-29.0	L_1
DG			515.3	-54.1	-109.1	-244.5	-14.0	
/	500	45	351.2	-39.6	-78.4	-177.8	-10.3	L_1
DG			176.2	-19.6	-38.9	-88.3	-5.1	
/	1000	30	164.9	-18.6	-36.7	-83.6	-4.8	L_1
DG			81.8	-9.2	-18.2	-41.6	-2.4	

Table 8

The simulation results under bus fault.

Grounded resistance/ Ω	Fault inception angle/ $^\circ$	k_1	k_2	k_3	k_4	k_5	Results
0	0	-362.9	-91.8	-181.7	-412.3	-23.8	Bus fault
10	90	-148.9	-36.4	-68.0	-162.2	-9.0	Bus fault
100	60	-428.9	-106.2	-212.2	-477.8	-27.5	Bus fault
500	45	-155.0	-39.3	-77.6	-176.5	-10.2	Bus fault
1000	30	-73.0	-18.6	-36.6	-83.7	-4.8	Bus fault

the amplitude of the zero sequence current [25].

If an SPG fault occurs on L_1 , the fault distance is 3 km, the fault inception angle is 30° , and the grounded resistance is 1000Ω . The comparison results are shown in Table 9.

From Table 9, we can see that the second faulty feeder selection method cannot select the faulty feeder accurately. Moreover, although the first method can determine the faulty feeder correctly, the power frequency component of the zero sequence current needs to be filtered out. In the simulation results of the third method, the difference of I_{m0} between L_1 and L_4 is not apparent, and the I_{m0} decreases with the deterioration of fault conditions, which is not conducive to selecting the faulty feeder. In contrast, when the proposed method is used for faulty feeder selection, the k of L_1 is opposite to the other four feeders, and the absolute value of the k is much larger. Accordingly, compared with the above three faulty feeder selection methods, the proposed method is easier to achieve good detection sensitivity in high-resistance grounded fault selection.

5. Field data test and analysis

In this paper, the proposed feeder selection method is verified by the field test data of an actual SPG fault in the resonant grounded system. The field test data are obtained from a substation with five feeders. The faulty feeder is chengxi feeder L_5 , and the other four feeders (Xuneng feeder L_1 , Xueyuan feeder L_2 , Rongjun feeder L_3 , and Taimeng feeder L_4) are healthy feeders. The sampling frequency of the wave-recording device is 12.5 kHz. The single-phase grounded fault occurred in L_5 of the Chengxi feeder at 15:27:18 on April 6, 2021. This paper simulates the field test data, the LWLR curves and their zero sequence current sloping curves are shown in Figs. 9 and 10, respectively.

According to Figs. 9 and 10, the slope value of L_5 at the fault time is positive, whereas the slope value of L_1 , L_2 , L_3 , and L_4 are negative. Therefore, there are apparent differences in the zero sequence current fitting curve and its sloping curve between L_5 and the healthy feeder.

Table 9

The faulty feeder selection results using different methods.

Faulty feeder selection method	Fault criterion	L_1	L_2	L_3	L_4	L_5	Results
Reactive power direction	Q_m	-12.215	22.765	60.342	194.13	5.7243	L_1
Distribution of zero sequence current	Sk	-0.0426	0.0001	0.0001	0.0001	-0.0001	/
Amplitude comparison	I_{m0}	10.1692	2.0667	3.9981	9.2618	0.5362	L_1
The proposed method	k	30.3	-3.9	-7.2	-17.3	-1.0	L_1

Field test data feeder selection results are given in Table 10.

According to the results in Table 10, the slope value k of the faulty feeder (Chengxi feeder L_5) is positive. In contrast, the slope values of the healthy feeders (L_1 , L_2 , L_3 , L_4) are negative. Consequently, based on the simulation analysis above, it is clear that the proposed method can detect the SPG fault accurately.

6. Conclusion

This paper analyzes the variation characteristics of the zero sequence current waveform of the SPG fault. Subsequently, an SPG faulty feeder selection method based on the slope of the zero sequence current fitting curve is proposed to solve the problem of low accuracy of the existing faulty feeder selection device. The main conclusions are as follows:

- (1) This method uses the slope value of the zero sequence current fitting curve to judge the faulty feeder, the principle is concise, and the calculation is simple.
- (2) This method can perform the LWLR on the sampled data to obtain its fitting curve directly because the power frequency component does not need to be filtered out. Therefore, combining the LWLR and sloping curve can simplify the calculation and obtain a good detection sensitivity in high-resistance grounded fault selection.
- (3) MATLAB/Simulink simulation results indicate that the proposed method can identify the SPG fault correctly under various fault conditions and keeps good detection sensitivity in high-resistance grounded fault selection.

CRedit authorship contribution statement

Aiyu Guo: Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing – original draft. **Xinhui Zhang:** Resources, Writing – original draft, Funding acquisition, Writing – review & editing. **Yu He:** Visualization, Investigation. **Shuyu Xie:** Resources,

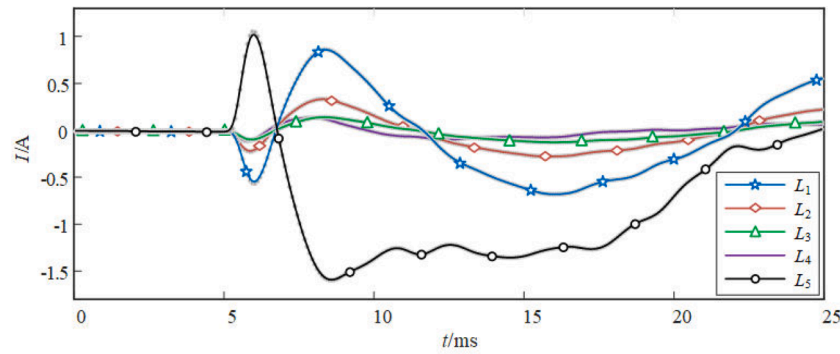


Fig. 9. The LWLR curves.

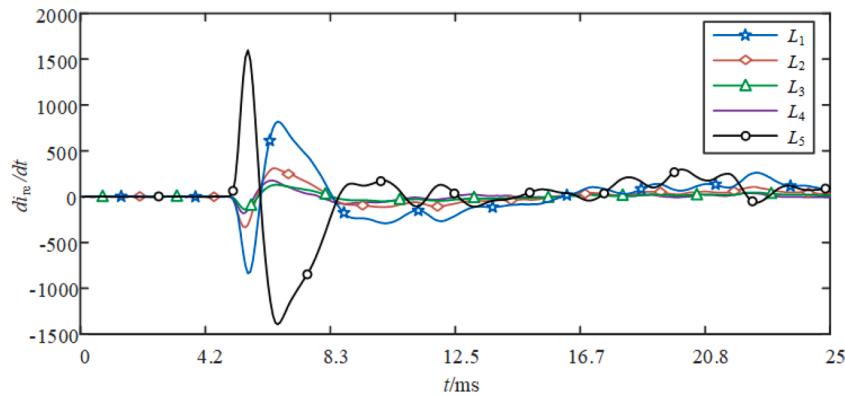


Fig. 10. The sloping curves.

Table 10

Faulty feeder selection results using Field data.

k_1	k_2	k_3	k_4	k_5	Results
-151.2	-267.5	-97.3	-283.2	709.9	L_5

Supervision. **Wenlong Li:** Software, Validation. **Haoyue Sun:** Visualization, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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